

Ala Dimassi

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📍 Monastir, Tunisia in LinkedIn 🌐 GitHub



Objective

Computer Science Engineering student passionate about Artificial Intelligence, Data Science, and Full-Stack Development. Experienced in building AI-powered systems using Machine Learning, LLMs, RAG architectures, and Business Intelligence tools.

Skills

Languages: Python, Java, JavaScript, SQL, C

AI & ML: XGBoost, SVM, scikit-learn, RAG, LangChain, ChromaDB, LLMs, Reinforcement Learning

Frameworks & Tools: FastAPI, Spring Boot, .NET, React, Next.js, Angular, Tailwind CSS, Power BI, Talend, n8n, Git, JavaFX

Domains: AI, Machine Learning, Business Intelligence, ETL Pipelines, Full-Stack Development

Experience

AI Engineering Intern(Summer Camp) – Talan Tunisie

Summer 2025

- Built an **AI-Powered Commercial Strategy Management Tool** combining LLMs, Reinforcement Learning, and RAG architectures within a team.
- Explored **multi-agent systems** and experimented with A2A and MCP protocols.
- Implemented **scraping and crawling pipelines** to gather, clean, and structure client and business data.
- Leveraged **MURAG techniques** to enhance contextual retrieval and improve RAG workflow performance.
- Presented and pitched technical work to stakeholders, communicating complex AI systems clearly.

Projects

Data Minds – AI Crypto Market Prediction Platform



Python, FastAPI, React, XGBoost, SVM, LangChain, ChromaDB, Ollama, Gemini

- Built AI models predicting Bitcoin/Ethereum direction using XGBoost and SVM; achieved **80% accuracy (BTC)** and **72% accuracy (ETH)**.
- Engineered 44+ indicators (RSI, MACD, EMA, SMA, ATR, Bollinger Bands) and developed a RAG assistant explaining predictions and feature importance.
- Integrated local and cloud LLMs using Ollama Llama 3.2 and Gemini 1.5 Flash.

Parapharmacie BI System (ESPRIT Integrated Project)



Power BI, Talend, n8n, Angular, FastAPI, LangChain, ChromaDB, LLMs

- Designed a **constellation schema data warehouse** (8 dimensions, 4 fact tables) covering Sales, Inventory, Payments, and Purchases for a Tunisian parapharmacie .
- Built **ETL pipelines with Talend** integrating ERP tables, CSV exports, and transaction logs into the data warehouse.
- Developed **Power BI dashboards** tracking KPIs across sales performance, stock levels, supplier reliability, and payment receivables.
- Implemented ML features: **RFM customer segmentation**, stockout prediction, product classification, and supplier recommendation.
- Built a **RAG assistant** using LangChain and ChromaDB enabling management to query business data in natural language.
- Automated data synchronization workflows using **n8n** and contributed to the **Angular** web interface.

Mental Health Support Platform (Hackathon – 6th/32 teams, 24h)



Next.js, Tailwind CSS, Bootstrap, JavaScript

- Built a platform connecting patients with psychologists for **online consultations**, real-time chat, and mental health progress tracking within 24 hours.
- Developed a fully responsive front-end using **Next.js** with **Tailwind CSS** and **Bootstrap**, covering booking flow, chat interface, and progress dashboard.
- Ranked **6th out of 32 teams** in a competitive hackathon environment.

Education

ESPRIT – École Supérieure Privée d'Ingénierie et de Technologies

Computer Science Engineering

2024 – Present

IPEIM – Institut Préparatoire aux Études d'Ingénieur de Monastir

Technology & Physics Preparatory Classes

2022 – 2024

Ranked **369/800** in the national engineering entrance exam.

Certifications

Introduction to Deep Learning & Neural Networks with Keras – IBM, Coursera

Machine Learning with Python – IBM, Coursera

Additional Information

Activities

- IEEE ESPRIT Student Branch
- Interact Club
- Notre Grand Bleu

Languages

- Arabic: Native
- French: B2
- English: B2